

Vannes WIJAYA

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SUMMARY

Final-year Computer Science student at Nanyang Technological University (QS #11) with proven expertise in low-level systems, algorithms and software engineering. Competed at international level in multiple IOI and ICPC, developing high-performance algorithmic mathematical solutions. Seeking full-time opportunities in Singapore, starting May 2027.

WORK EXPERIENCE

Huawei – Machine Learning Engineer

Aug 2026 – Present

- Developing computer-use agents for OpenJiuwen, including experimental results analysis, hyperparameter tuning, prompting, memory management, alignment, multi-modal reasoning and planning
- *Skills: Python, Agentic Coding, Harness Design*

Ader Co – Fullstack Developer Freelance

Jan 2026 – Mar 2026

- Developed AI-assisted online learning service web application, using Next.js and PostgreSQL
- Managed end-to-end development process, from system design, backend, frontend, authentication to deployment
- *Skills: Next.js, React.js, PostgreSQL, Web Development*

Huawei – Processing Algorithm Engineer

Aug 2025 – Dec 2025

- Developed PyPTO, python framework for Huawei NPU to optimize for developer's ease-of-use, runtime efficiency, and numerical accuracy
- Utilized internal framework to implement and deploy state-of-the-art LLMs and machine learning algorithms
- Delegated to Huawei Nanjing Research Center in China to integrate the framework and models for high-performance computing pipelines
- *Skills: GPU Programming, Ascend & CANN, Python, C++, CMake, Docker, Linux*

ST Engineering – Machine Learning Engineer

May 2025 – Aug 2025

- Developed production-grade ML pipelines utilizing LLMs, VLMs, and object detection models, using speculative decoding to improve efficiency of internal libraries
- Engineered service integration for third-party Amazon devices
- *Skills: LLM, VLM, YOLO, Computer Vision, Data Pipeline, Python, Node.js, Git, Docker, Linux*

EDUCATION

Nanyang Technological University – Computer Science (Bachelor)

2023 – Present

- Expected graduation in May 2027 with First Class Honours (GPA 4.84 / 5.00)
- Dean's List Award Academic Year 2024–2025
- GEM Student Exchange Programme 2026 to Sogang University, South Korea

TECHNICAL SKILLS

Programming: Python, C/C++, Java, Javascript, TypeScript, SQL

Machine Learning: PyTorch, LangChain, Numpy, Scikit-learn, Pandas, Matplotlib

Web: React.js, Next.js, Node.js, Express.js, MongoDB, PostgreSQL, TailwindCSS

Tools: Git, Docker, Linux, CMake, AWS S3

Languages: English (Professional), Indonesian (Native), Chinese (Basic), Japanese (Basic)

ACHIEVEMENTS

- **ICPC Asia Pacific Championships** – 1 Bronze Medal, 1 Honorable Mention
- **ICPC Regionals** – 1 Gold Medal, 3 Silver Medals
- **International Olympiad in Informatics (IOI)** – 2 Silver Medals
- **Ruangguru Road of Champions** – Runner Up
- **Ruangguru Clash of Champions II** – Winner
- **TCS CodeVita** – Second Runner Up
- **Google Kickstart** – Global Rank Top 48th

PROJECTS

NTU Final Year Project on Deep Evolving Network

Present

- Ongoing research on Explainable AI methods and neuro-fuzzy architectures, as part of NTU Final Year Project.

PYPHub Web Forum

2025

- Developed student forum for past-year-paper discussion and automated solver, as part of Students' Computing and Data Science Club initiatives
- Engineered data ingestion pipeline to preprocess and upload papers and course materials to AWS S3 and OpenAI vector stores, significantly improving resource accessibility for both users and LLMs in solution generation

NTU URECA Undergraduate Research Programme

2025

- Designed and evaluated deep learning agents to solve classic optimization and decision problems, using methods inspired from AlphaGo-style self-play training

AI & ML Club Website

2025

- Developed official club page, as part of NTU Artificial Intelligence and Machine Learning Club initiatives
- Developed responsive website using React.js, TypeScript and TailwindCSS, leveraging open-source libraries and design systems for enhanced functionality and user interfaces

ParkNGo Web Application

2024

- Developed full-stack web app with realtime search, filtering and geolocation features for car park management and discovery, as part of Software Engineering Coursework. Stack includes React.js, Next.js, and TailwindCSS

TOKI Open Contest Problemsetting

2023

- Curated high-quality algorithmic problems, full-coverage test suites, and detailed editorials for a competitive programming contest with hundreds of participants across the globe
- Problems are used as part of TOKI Regular Open Contest 31, sponsored by official Indonesia IOI Training Team